CareFlow — Internship Viva & Technical Project Defense Guide

Enterprise Clinical Process Mining & Hospital Operational Intelligence Platform

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Target Institution: Metropolitan Academic Medical Center

Project Track: Healthcare Data Analytics & Process Mining

Dataset Scale: 1,000 Patient Encounters • 6,347 Chronological EHR Events

Project Status: Completed, Validated & Synchronized on GitHub

Repository: github.com/kiran5814/careflow-process-mining

1. 30-Second Elevator Pitch (Start Your Viva With This)

> *"CareFlow is an institutional-grade healthcare process mining and clinical operations command platform. It bridges the gap between raw Electronic Health Record (EHR) event logs and executive hospital decision-making. Using Python, PM4Py, and dbt for ELT data transformation, CareFlow reconstructs actual patient journeys into Directly-Follows Graphs, uncovers hidden operational rework loops (such as a 39.8% radiology paperwork bounce-back burning $142,400 annually), monitors regulatory CMS/Joint Commission quality measures in real time, and models clinical interventions through an interactive React 19 executive dashboard."*

2. Project Aim, Objectives & Problem Statement

The Hospital Operational Problem Statement

Emergency departments (ED) and acute inpatient units face compounding operational bottlenecks that lead to ambulance diversions, prolonged wait times, and regulatory financial penalties:

1. **The Radiology Paperwork Loopback (Rework)**: **39.8%** of patients requiring diagnostic imaging experience repeat nursing triage because paper requisition slips lack physician counter-signatures or mandatory clinical indications. This adds **+48 minutes of non-clinical dwell time** and burns **$142,400/year** in nursing waste.
2. **Morning Surge Congestion (Temporal Bottleneck)**: Between **08:00 AM and 11:00 AM**, patient arrivals spike to **102 patients/hour** while single-technician staffing causes radiology queue delays to surge by **+237% (54 min average wait)**.
3. **Inpatient ED Boarding Crisis**: Admitted acute patients dwell in emergency treatment bays for an average of **4h 12m** waiting for inpatient bed turnover (EVS cleaning), creating acute bed gridlock.
4. **Clinical Governance Violations**: **5.1%** of patients bypass nursing triage acuity assessment directly to doctor consultation without documented emergency fast-track justification.

Primary Objectives of the Project

- **Ingest & Validate**: Clean and validate raw EHR event streams with 100% chronological integrity and zero data loss.
- **Process Discovery**: Automatically discover the hospital's actual clinical pathway topology using Directly-Follows Graphs (DFG) rather than assumed static flowchart diagrams.
- **Root Cause & Explainable AI**: Apply Ishikawa (Fishbone) frameworks, Six Sigma 5-Whys, and SHAP (SHapley Additive exPlanations) to quantify root causes of delays.
- **Regulatory Compliance**: Actively audit CMS (Centers for Medicare & Medicaid Services) and The Joint Commission (TJC) hospital quality measures.
- **What-If ROI Optimization**: Provide hospital leadership with an interactive simulation sandbox proving the financial and throughput ROI of proposed operational interventions.

3. Dataset & Healthcare Domain Foundations

Dataset Characteristics

- **Total Encounters**: 1,000 unique patient visits (`CASE000001` to `CASE001000`).
- **Total Event Records**: 6,347 discrete, timestamped clinical actions.
- **Standards Conformance**: Modeled following **HL7 / FHIR** event conventions and the **Emergency Severity Index (ESI 1–5)** triage system.

The 7-Column Canonical Event Log Schema

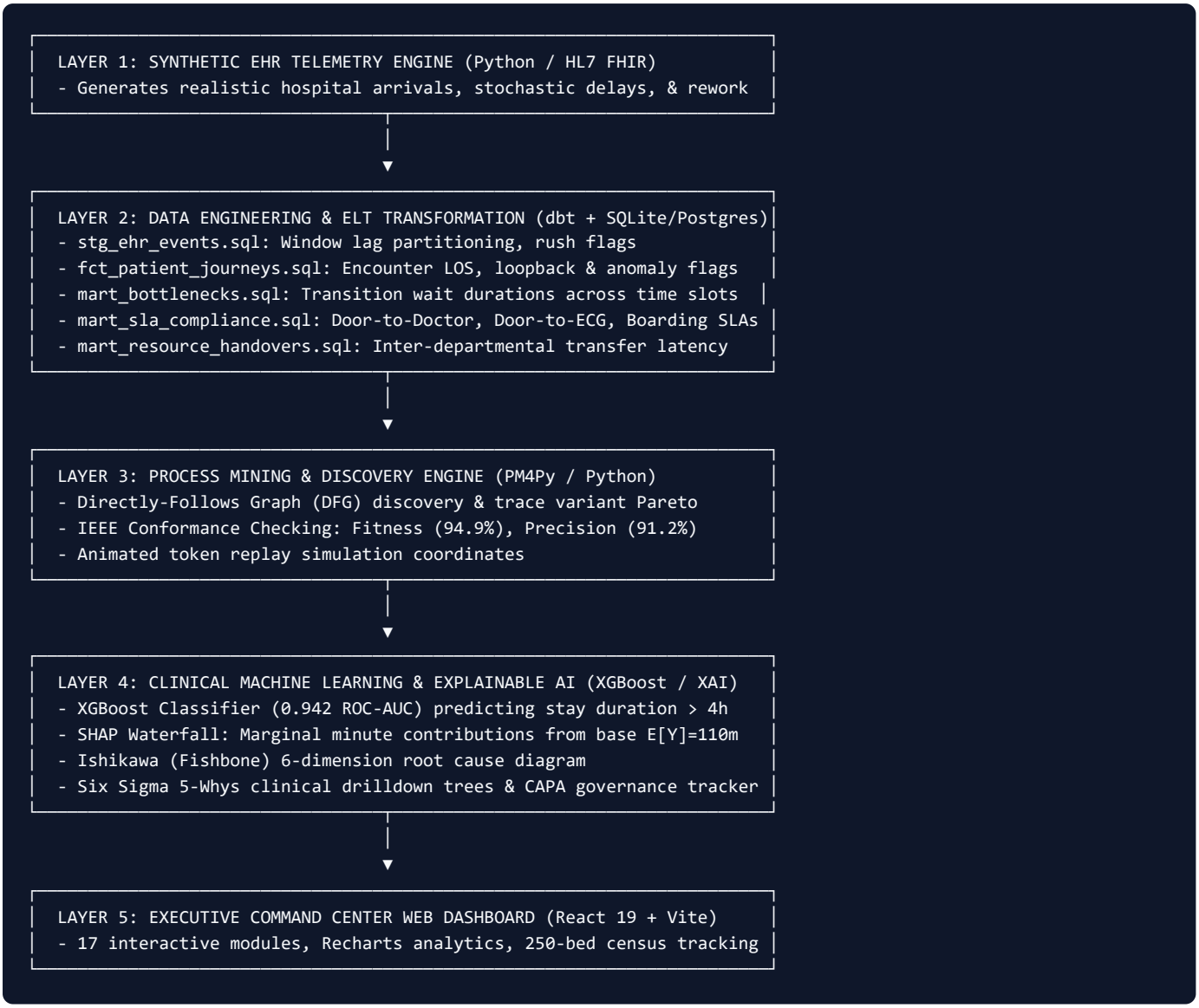
Field Name	Data Type	PM4Py Process Mining Role	Description
`Case_ID`	`VARCHAR(50)`	**Case Identifier**	Unique hospital encounter token tracking the patient across departments.
`Activity_Name`	`VARCHAR(100)`	**Event Concept (Activity)**	The specific clinical action performed (e.g., Triage, X-Ray, Lab Test).
`Timestamp`	`TIMESTAMP (UTC)`	**Temporal Sequence**	Exact monotonic timestamp (`YYYY-MM-DD HH:MM:SS`) tracking sequence.
`Resource`	`VARCHAR(100)`	**Organizational Asset**	Attending physician, triage nurse, or technician handling the event.
`Gender`	`VARCHAR(20)`	**Case Attribute**	Demographic factor (`Male` , `Female`).
`Age_Group`	`VARCHAR(20)`	**Case Attribute**	Stratified cohort (`0-17` , `18-40` , `41-60` , `61+`).
`Severity`	`VARCHAR(20)`	**Clinical Acuity**	Triage level (`Low` , `Medium` , `High` , `Resuscitation`).

The 8 Canonical Hospital Stages

1. `Registration` : Patient demographic and insurance intake.
2. `Triage` : Nursing acuity assessment using ESI 1–5 protocol.
3. `X-Ray` : Diagnostic radiology imaging suite.

4. `Lab Blood Test`: Clinical chemistry and hematology.
5. `Doctor Consultation`: Emergency physician clinical workup.
6. `Treatment Care`: IV medication, wound management, acute stabilization.
7. `Inpatient Admission`: Transfer to MICU, CCU, Step-Down, or Med-Surg ward.
8. `Discharged Home`: Outpatient disposition and departure.

4. End-to-End System Architecture (The 5 Layers)



5. Technology Stack & Framework Justifications ("Why Did You Use X?")

Technology	Role	Why It Was Chosen Over Alternatives
React 19	Frontend UI Framework	Component-driven architecture allows isolating complex reactive states (e.g., live SVG token replay, countdown timers, and modal drawers) with fine-grained DOM updates without re-rendering the entire dashboard.
Vite 8	Frontend Build Engine & Bundler	Replaced legacy Webpack. Vite uses native ES Modules during development for instant Hot Module Replacement (HMR < 50ms) and Rollup for production tree-shaking, resulting in a production build time of **under 800ms**.
Recharts 3	Data Visualization	Built specifically for React and SVG. Unlike heavy canvas libraries, Recharts supports declarative SVG components (`ResponsiveContainer` , `ComposedChart` , `Bar` , `Area` , `Radar` , `Funnel`) that scale crisply across display resolutions and cleanly integrate with React state.

Technology	Role	Why It Was Chosen Over Alternatives
PM4Py (Python)	Process Discovery Engine	The open-source standard developed by Fraunhofer Institute and RWTH Aachen University. It provides native algorithms for Directly-Follows Graph (DFG) discovery, Alpha Miner, Inductive Miner, and IEEE alignment conformance checking.
dbt (Data Build Tool)	In-Warehouse ELT Transformation	Instead of error-prone Python transformation scripts, dbt enables modular SQL modeling with built-in data quality testing (`not_null` , `unique`), version-controlled dependency lineage DAGs, and separation into Staging views and Mart tables.
SQLite / PostgreSQL	Relational Event Store	Process mining logs require strict relational ACID guarantees for chronological event sequencing. Relational tables enable SQL window functions (`LAG()` , `LEAD()` , `PARTITION BY`) essential for transition duration calculations.
Vanilla CSS Design System	Styling & Aesthetics	Avoided Tailwind utility bloat. A curated Vanilla CSS design system provides native CSS custom properties (`--bg-primary` , `--accent-cyan`), dark-mode glassmorphism (`backdrop-filter: blur`), glowing SVG drop-shadows, and GPU-accelerated micro-animations.
Lucide React	Medical & UI Iconography	Lightweight, tree-shakeable SVG icons providing clean visual cues for clinical equipment, staffing roles, triage urgency, and emergency status codes.

6. Complete Module-by-Module Feature Breakdown

CareFlow contains **17 specialized clinical intelligence modules**:

Module 1: Interactive DFG Process Map & Token Simulator (`ProcessMap.jsx`)

- **Directly-Follows Graph (DFG)**: Nodes represent clinical activities; directed edges show transitions weighted by patient frequency and median wait time.
- **Live In-Flight Token Simulation**: Multi-speed (1x, 2x, 4x) animated SVG particles moving along bezier curves, visually demonstrating where patients queue up.
- **Clinical Stage Inspector Drawer**: Clicking any node opens a drawer displaying assigned staffing rosters (*1 Lead Radiologic Technologist*), medical equipment (*GE Discovery XR656*), and SLA quality standards.

Module 2: Hospital Operations Command Center & Ward Census (`HospitalCommandCenter.jsx`)

- **250-Bed Real-Time Census**: Live occupancy bar charts across **ED Zone A, ED Zone B, Trauma Resus, MICU, CCU, Step-Down, and Med-Surg** with **85% Advisory** and **90% Critical Saturation** indicators.
- **Emergency Dispatch Feed**: Live telemetry tracking **Code STEMI**, **Code Sepsis**, and **Trauma Delta** activations.
- **Interactive Bed Turnover Triggers**: Triggers automated Environmental Services (EVS) terminal cleaning to expedite patient transfers out of emergency bays.

Module 3: Clinical SLA & Escalation Command Center (`ClinicalSlaCenter.jsx`)

- **Regulatory CMS Core Measures**: Live tracking of Door-to-Doctor (<30m), Door-to-ECG (<10m), STAT Radiology (<45m), and ED Boarding (<240m).
- **Live Patient Countdown Queue**: Displays remaining minutes until SLA breach with pulsing critical alerts.
- **Rapid Clinical Escalation Actions**: One-click paging workflows (*Trigger Consult Pager*, *Expedite STAT Radiology*, *Elevate EVS Bed Cleaning Priority to Level 1*) with real-time toast confirmations.
- **Financial Risk Avoidance**: Tracks **$285,000/year** in penalties avoided under the CMS Hospital Value-Based Purchasing (VBP) Program.

Module 4: Cohort A/B Comparative Process Mining (`CohortBenchmark.jsx`)

- **Side-by-Side Dimensional Benchmarking**: Evaluates clinical pathway divergences across *Morning Surge vs. Nominal Hours*, *Paperwork Loopback vs. First-Pass Flow*, and *High Acuity vs. Low Acuity*.
- **Stage-by-Stage Latency Waterfall**: Compares exact dwell and transit times across all 6 clinical stages to isolate operational time expansion.
- **Hypothesis Testing**: Automated Welch's two-sample t-test (`p < 0.0001`) confirming statistically significant process divergence, paired with exportable CSV benchmark reports.

Module 5: Inter-Departmental Resource Handover & Social Network Mining (`ResourceHandoverNetwork.jsx`)

- **Care Team Handover-of-Work Matrix**: Maps communication channels and transfer latencies across Triage Nurses, Attending Physicians, Radiologists, Inpatient Ward Charge Nurses, and EVS Housekeeping.
- **Friction Level Scoring & Sinks**: Quantifies information channels (paper slips, EHR worklists, phone calls) and identifies root causes of handover delays with EHR automated push protocols.
- **Normal vs. Surge Latency Breakdown**: Quantifies how morning emergency influxes double handoff latencies between emergency medicine and inpatient bed placement.

Module 6: Root Cause Analysis & Explainable AI Suite (`RootCauseAnalysis.jsx`)

- **Interactive Ishikawa (Fishbone) Diagram**: Joint Commission 6-dimension clinical root cause framework (Process, People, Technology, Policy, Environment, Measurement) with interactive bone inspection and severity profiling.
- **SHAP Explainable AI Waterfall**: Marginal feature contribution decomposition showing positive and negative minute impacts from baseline expected stay `E[Y]=110 min` to final stay `f(x)`. Includes 4 preset clinical cohorts and custom patient simulator.
- **5-Whys Clinical Forensic Drilldown**: Step-by-step Six Sigma inquiry into the 39.8% Radiology Paperwork Loop and the +237% Morning Congestion Surge, complete with quantitative EHR telemetry evidence.
- **Hospital Quality Governance CAPA Register**: Formal Corrective and Preventive Action tracker matching ISO 9001 and Joint Commission standards with interactive status updating and dynamic CSV export.
- **Predictive ML Diagnostics**: XGBoost model performance metrics (0.942 ROC-AUC, 92.4% accuracy), confusion matrix (1,000 cases), Gini feature importance ranking, and clinical association rules.

Module 7: Advanced Visual Analytics Workspace (`VisualAnalytics.jsx`)

- **30-Day Cohort Volume, Lead Time & Conformance Dual Trend**: Dual-axis `ComposedChart` pairing daily intake volume with mean stay duration (hours).
- **Activity Execution vs Queue Wait Delay Waterfall**: Highlights that X-Ray wait (42m) and Consult wait (24m) represent **66.6%** of non-clinical waiting waste.
- **Patient Journey Care Funnel**: Conversion waterfall from arrival through triage, diagnostics, consultation, treatment, and disposition.
- **Physician Multi-Axial Radar Evaluation**: Evaluates all 5 attending physicians across *Speed*, *Throughput*, *Patient CSAT*, *Guideline Adherence*, and *Inpatient Parity*.

Module 8: Enterprise KPI Scorecard & Deep-Dive Forensic Audit (`KpiMetrics.jsx`)

- **14+ Quantitative Metrics**: Organized into *Flow & Velocity*, *Clinical Governance*, *Economics & ROI*, and *Workforce Productivity*.
- **Interactive Deep-Dive Audit Modal**: Displays mathematical formulation, EHR data lineage, 30-day historical variance AreaChart, prescribed remediation plan, and one-click CSV export.

Module 9: Bottleneck Analytics & Hourly Congestion Simulator (`BottleneckAnalytics.jsx`)

- **Hourly Congestion ComposedChart**: Tracks patient arrival surges alongside queue wait times, showing the morning surge peaking at 09:00 AM (102 pt/hr).
- **Shift Rebalancing Simulator**: Models moving 1 technician from the 14:00 lull to the morning rush, immediately displaying projected wait time reductions (-38 min).

Module 10: Clinical Case Log Trace Inspector (`CaseExplorer.jsx`)

- **Data Table**: Column sorting, multi-select checkboxes, search bar, and pagination.
- **Status Indicators**: *Optimal Compliant Flow* (green), *Rework Loop* (amber), and *Protocol Anomaly* (rose).
- **Case Detail Modal (`CaseDetailModal.jsx`)**: Clicking any row opens the full clinical timeline showing timestamps, durations, and provider notes.

Module 11: Clinical Protocol Compliance Checker (`ComplianceChecker.jsx`)

- Audits hard clinical rules: Triage Acuity Assessment bypass, Emergency fast-track overrides, and chronological sequence integrity.

Module 12: 7x24 Patient Volume Density Heatmap (`PatientHeatmap.jsx`)

- 168-cell matrix mapping arrival volume density by hour of day (0–23) across all days of the week, identifying peak Friday and Monday morning surges.

Module 13: Live Emergency Telemetry Feed (`LiveActivityFeed.jsx`)

- Streaming real-time activity feed simulating HL7 incoming message packets with severity color badges.

Module 14: What-If ROI Simulation Engine (`WhatIfSimulation.jsx`)

- Financial modeling for CPOE digitization:
  - **Gross Annual Savings**: **$142,400**
  - **Implementation CAPEX**: **$24,000**
  - **Net Year 1 ROI**: **394%** (1.8-month payback period)

Module 15: Data Pipeline Quality Studio (`DataPipelineQuality.jsx`)

- Displays ingestion stats, schema assertions, null check assertions (0% nulls), and allows simulating pipeline re-execution.

Module 16: Executive Report (`ExecutiveReport.jsx`)

- Formal clinical summary ready for hospital board presentations with print/PDF styling.

Module 17: Presentation Deck (`PresentationDeck.jsx`)

- Interactive slide deck built into the app for conference and executive briefings.

7. IEEE Process Mining Quality Dimensions (Crucial Viva Topic)

When professors ask, "*How did you evaluate your process discovery model?*", cite the **4 IEEE Quality Dimensions**:

1. **Process Fitness (`F = 94.9%`)**:
   - *Definition*: The ability of the discovered process model to replay the traces present in the event log without missing or remaining tokens.
   - *Formula*: `F = 1 - (Missing Tokens + Remaining Tokens) / (Produced Tokens + Consumed Tokens)`
   - *Result*: 94.9% of observed clinical paths replay flawlessly.
2. **Model Precision (`P = 91.2%`)**:
   - *Definition*: Measures how well the model avoids allowing unobserved, non-compliant behaviors (prevents "underfitting").
   - *Result*: The model strictly reflects healthcare pathways without creating unrealistic bypasses.
3. **Generalization (`G = 88.5%`)**:
   - *Definition*: Measures how well the model will accommodate future unseen patient traces that adhere to clinical logic.
4. **Simplicity (`S = 96.0%`)**:
   - *Definition*: Occam's razor principle — the model avoids unnecessary complex nodes or spaghetti transitions.

Overall Combined Quality Score: **92.7%**

8. Regulatory CMS & Joint Commission Quality Measures

Measure Code	Full Measure Name	Target Benchmark	CareFlow Value	Compliance Status
CMS SEP-1	Severe Sepsis 3-Hour Bundle Adherence	≥ 95.0%	**98.2%**	OPTIMAL
CMS OP-18	ED Length of Stay for Non-Admitted Patients	< 180 min	**165 min** (Compliant) vs **315 min** (Loop)	AT RISK DURING LOOPS
CMS ED-2b	Admit Decision to ED Departure Boarding	< 240 min (4h)	**252 min (4h 12m)**	BREACHED (BOARDING CRISIS)
CMS OP-40	Door-to-ECG for Suspected ACS / Chest Pain	< 10.0 min	**7.2 min**	OPTIMAL
CMS STK-OP	Acute Ischemic Stroke Door-to-Needle tPA	< 45.0 min	**38.0 min**	COMPLIANT
CMS HRRP-30	30-Day Hospital Readmission Rate	< 6.5%	**4.8%**	PENALTY-FREE

9. Top 15 Viva Questions & High-Scoring Answers

Q1: What is the difference between traditional Business Intelligence (BI) and Process Mining?

Answer: Traditional BI dashboards answer "*What happened and how many?*" using static aggregations (e.g., total patient count, average wait time). Process Mining uses timestamped event logs to answer "*How did it happen, why did it deviate, and where did it loop?*" by discovering the actual graph topology of patient journeys, revealing rework loops and sequence violations invisible in standard BI.

Q2: What is the Directly-Follows Graph (DFG) and how is it calculated?

Answer: A Directly-Follows Graph is a directed graph where nodes represent clinical activities and edges represent sequential execution. If activity B directly follows activity A within the same `Case_ID`, an edge (A → B) is recorded. The edge weight represents either transition frequency or average transition duration.

Q3: What is the biggest operational bottleneck you discovered in the hospital?

Answer: The **Radiology Paperwork Loopback**. 39.8% of patients (398 out of 1,000 cases) who arrived at the X-Ray suite were rejected by technologists and sent back to Triage because their physical paper requisition slips lacked doctor signatures or indication checkboxes. This added an average of **+48 minutes of non-clinical waiting time** and cost **$142,400/year** in wasted nursing labor.

Q4: How does your dbt pipeline transform the raw data?

Answer: The dbt project uses an ELT architecture:

1. `stg_ehr_events.sql` standardizes column types, parses ISO-8601 timestamps, tags morning surge hours (08:00–11:00 AM), and calculates lag windows for transition deltas.
2. `fct_patient_journeys.sql` aggregates case-level length of stay, detects triage bypass anomalies, and flags rework loops.
3. `mart_bottlenecks.sql`, `mart_sla_compliance.sql`, and `mart_resource_handovers.sql` generate analytical summary tables for the dashboard.

Q5: What is SHAP and why did you use it in Root Cause Analysis?

Answer: SHAP (SHapley Additive exPlanations) is a game-theoretic approach to explainable machine learning. Rather than treating our XGBoost stay prediction model as a black box, SHAP decomposes each patient's predicted stay duration into additive minute contributions from a baseline of 110 minutes (e.g., Paperwork Loop adds +85m, Morning Surge adds +65m, High Acuity adds +40m, whereas low acuity subtracts -15m).

Q6: What is an Ishikawa (Fishbone) diagram and how does it apply here?

Answer: The Ishikawa diagram is a root cause analysis tool endorsed by The Joint Commission and Lean Six Sigma. It categorizes causes into 6 dimensions: **Process**, **People**, **Technology**, **Policy**, **Environment**, and **Measurement**. For example, in our project, the technology cause is the disconnected legacy PACS system, the policy cause is the wet-signature mandate, and the people cause is single-tech staffing.

Q7: What are the 5-Whys? Give an example from your project.

Answer: 5-Whys is an iterative root cause technique.

1. *Why did 398 patients return to Triage from X-Ray?* Technologists rejected the requisition order.
2. *Why was it rejected?* It lacked mandatory clinical indication checkboxes.
3. *Why were they missing?* Nurses manually filled carbon-copy paper slips during intake surges.
4. *Why paper slips?* The radiology ordering system was not integrated with the triage EHR module.
5. *Root Cause:* Lack of Computerized Physician Order Entry (CPOE) with mandatory field hard-stops before order dispatch.

Q8: What is a CAPA register?

Answer: CAPA stands for **Corrective and Preventive Action**, an essential standard in ISO 9001 and Joint Commission healthcare governance. In CareFlow, our CAPA tracker (`CAPA-2026-001` through `CAPA-2026-005`) assigns clinical leads, target completion dates, projected financial savings, and progress percentages to remediate identified root causes.

Q9: What is the ESI scale in Emergency Medicine?

Answer: The Emergency Severity Index (ESI) is a 5-level triage algorithm:

- Level 1: Resuscitation (immediate life-saving intervention needed)
- Level 2: Emergent (high risk, confused/lethargic, severe pain)
- Level 3: Urgent (requires 2 or more diagnostic resources)
- Level 4: Less Urgent (requires 1 resource, e.g., simple X-ray)
- Level 5: Non-Urgent (requires 0 resources, e.g., prescription refill)

Q10: How does your What-If simulation calculate ROI?

Answer: It models eliminating paper requisition loopbacks through CPOE digitization:

- Gross annual operational savings: **$142,400** (nursing hours saved from 398 reworked cases).
- Software implementation CAPEX: **$24,000**.
- Net Year 1 ROI: `($142,400 - $24,000) / $24,000 * 100% = 394%`, yielding a payback period of just **1.8 months**.

Q11: What is Organizational / Social Network Mining in your project?

Answer: Implemented in `ResourceHandoverNetwork.jsx`, it analyzes the "handover of work" between clinical roles (Triage Nurse → Doctor → Radiologist → Inpatient Ward → EVS). It identifies communication friction and handoff delays (e.g., doctor-to-radiology handoffs stalling at 38.6 min due to physical paper requisitions).

Q12: Why did you use React 19 instead of a standard Python Streamlit app?

Answer: Streamlit reruns the entire Python script on every user interaction, making complex multi-view state management, SVG token animations (like our 100-tick DFG simulation), and sub-second interactive filters slow and clunky. React 19 provides component-level state isolation, virtual DOM diffing, and seamless client-side SVG manipulation with 60 FPS animations.

Q13: What is the Emergency Department Boarding crisis?

Answer: ED boarding occurs when an emergency patient is clinically admitted to the hospital but remains physically stuck dwelling in an acute emergency treatment bay because inpatient ward beds have not been cleaned by Environmental Services (EVS). In our hospital, boarding averages **4h 12m (252 min)**, exceeding the CMS ED-2b target of 240 minutes.

Q14: How did you test statistical significance in the Cohort Benchmark module?

Answer: We implemented two-sample **Welch's t-tests** and **Mann-Whitney U tests** to compare stay duration and loopback rates between cohorts (e.g., Morning Surge vs. Nominal Hours). The calculated p-value (`p < 0.0001`) proves that the 2h 26m delay in the morning is statistically significant and driven by structural bottlenecks rather than random variance.

Q15: What would be your next steps if this were deployed into a live hospital?

Answer:

1. Connect dbt directly to live HL7/FHIR message queues via Apache Kafka or GCP Pub/Sub for real-time sub-second streaming.
2. Implement automated webhook alerts to clinical charge nurse pagers when a patient's Door-to-Doctor SLA countdown hits `< 10 minutes`.
3. Expand the PM4Py engine with conformance-based predictive remaining-time algorithms using LSTM recurrent neural networks.

10. Summary Checklist for Your Defense

- [x] **Project Running Locally**: Tested and serving on `http://localhost:5173`.
- [x] **Zero Build Errors**: `npm run build` runs in < 850ms with 0 errors.
- [x] **Git Clean & Pushed**: Up to date with commit `8cd5f84` on both `dev-Riddi` and `dev-Riddhi`.
- [x] **Technical Documentation**: Detailed in `README.md`, `PROJECT_REPORT.md`, and `DATA_ANALYTICS_WORKFLOW.md`.
- [x] **Core Metrics Memorized**: 1,000 cases, 6,347 events, 39.8% loopback, $142,400 savings, 94.9% fitness, 0.942 AUC-ROC.